

CODESPLASH

SCHOOL PHASE

VÉQALUNE CIVIC

AI-Powered Community Issue Intelligence Platform

Theme 01: Smart Cities & Sustainable Communities

SVD Synapse **Ananda College, Colombo 10**

M. Sinura Damsath Perera (Team Leader)

K.A. Vihas Dinsitha Kodithuwakku

J.M.G.W. Dulnith Sanvidu Bandara

Contact: +94 77 447 0098, sinuradperera@gmail.com

Live Platform: <https://veqalune-civic.vercel.app>

Brand Identity:

<https://drive.google.com/drive/folders/1qPBbFLikWXxCNEQWPWH93iG3dF1xvIB2?usp=sharing>

Promotional Video:

https://drive.google.com/file/d/1sKDZtQWStPz2W2M4r6djzpRse_XJ8IEa/view?usp=sharing

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1. Executive Summary

VÉQALUNE CIVIC is an AI-powered platform built on one simple idea: a civic complaint should not simply vanish into a phone call, a message, or a social media post. This should be useful information which helps a local authority to see what is going on and where it is going on and what needs attention first. This proposal addresses Theme 01: Smart Cities & Sustainable Communities by dealing with the everyday issues that affect the quality and safety of community life.

In many urban and semi-urban areas in Sri Lanka, people still use separate channels to report problems such as phone calls, walk-in complaints, informal messages and social media. These channels can alert an authority that something is wrong, but they rarely give a clear picture of how severe the issue is, where exactly it is, whether it keeps happening, or what its current status is. This creates a real problem for officers who already have limited time and tight maintenance budgets. They may know that dozens of complaints exist, but not which ones need attention right now, or which repeated complaints point to a bigger problem underneath.

VÉQALUNE CIVIC turns a citizen report into structured decision-support data. A citizen can submit a photo, a short description, and a location. The platform then uses Google Gemini multimodal analysis to examine the image and text together. It identifies the likely issue category, estimates severity, considers environmental and public-safety risk, assigns a confidence level, and provides a plain-language explanation. A transparent 0–100 Priority Score is then calculated using these factors together with location sensitivity and recurrence. The goal is not to replace the judgement of municipal officers. The goal is to give them a clearer and more consistent starting point.

The platform also looks beyond individual complaints. Nearby and repeated reports can be grouped into geographic hotspots, while the command dashboard provides a quick view of active, critical, high-priority, and resolved issues. The insight and prediction modules then highlight recurring patterns and possible future risks, such as drainage trouble before heavy rainfall.

The working prototype uses React, TypeScript, Vite, Node.js, Express, Leaflet, and the Gemini API. A deterministic rules-based scoring engine is included as a fallback, so the core prioritisation workflow can still operate when live AI access is unavailable. This combination gives the project both innovation and practical resilience. Most importantly, the system is designed to be understandable to real people making real operational decisions.

One of the key advantages of VÉQALUNE CIVIC is its multilingual accessibility, allowing users across Sri Lanka to interact with the platform in their preferred language. The application supports all three official languages of Sri Lanka—Sinhala, Tamil, and English—ensuring that language barriers do not prevent citizens from reporting community issues, accessing important information, and engaging with civic services. This inclusive approach makes the platform more accessible to a wider range of communities and helps ensure that everyone has an equal opportunity to make their voice heard.

2. Problem Statement

2.1 Problem Description

Local government authorities across Sri Lanka deal with a large number of small but important infrastructure and public-service problems every day. These include overflowing waste bins, construction waste dumping, potholes and damaged roads, blocked drains, water leaks, electrical hazards, broken streetlights, and public-safety risks. Each issue may look small when reported on its own. The bigger challenge appears when many such issues arrive through disconnected channels and there is no shared way to compare them.

A complaint recorded only as a phone message or social media post may have little structured information attached to it. The location may be vague, the severity may be unclear, and there may be no easy way to connect the report to earlier complaints from the same area. As a result, responses can become first-come-first-served or ad hoc. An issue that creates a real safety or environmental risk may wait behind a less urgent complaint simply because the latter arrived earlier or was easier to understand.

2.2 Background and Context

Urban councils and Pradeshiya Sabhas operate with limited staff, time, and budgets. For this reason, simple operational questions matter. Which locations are repeatedly affected by waste dumping? Which road section keeps generating damage reports? Which drain becomes a concern before heavy rain? Which issue categories are creating the highest number of unresolved reports? Without structured information, answering these questions requires manual checking across many separate records. Citizens face the problem from the other side. A person may take the time to report an issue but receive little visibility afterward. They may not know whether the report was received, whether it was considered urgent, or whether another person has already reported the same problem. Over time, this weak feedback loop can reduce willingness to participate. A better system needs to support both sides: citizens need a simple way to report, while officials need a clear way to understand and act.

2.3 Importance of the Problem

- Delayed response can turn a manageable problem into a serious public-safety or environmental incident, especially when exposed electrical equipment, major road damage, or blocked drainage is involved.
- Repeated complaints may remain symptoms instead of becoming evidence of a hotspot, so authorities repeatedly patch the same problem without addressing the underlying cause.
- Maintenance budgets can be spread too thinly when decisions are made reactively rather than according to risk, recurrence, and likely community impact.
- Citizens are less likely to keep reporting problems when they see no clear acknowledgement or follow-up.

VÉQALUNE CIVIC addresses these gaps by creating one shared view of civic issues. The system does not simply collect complaints. It helps turn them into structured evidence that can support prioritisation, planning, and longer-term community improvement.

3. Proposed Technical Solution

3.1 Solution Overview

VÉQUALUNE CIVIC is a full-stack web platform built around three connected workflows. First, a citizen submits an issue with a photograph, a short description, a category, and a location. Second, the platform analyses the submission and turns it into structured information. Third, municipal users can view individual reports together with hotspots, dashboards, and higher-level insights. These workflows are connected so that information collected at the reporting stage remains useful for decision-making later. The prototype is intentionally practical. Citizens do not need to understand AI, GIS, or scoring formulas, and officials do not need to inspect raw model output. Each stage turns complex processing into information that can be understood quickly, supporting both usability and transparency.

3.2 Key Features

- Guided citizen reporting: a user can capture or upload a photo, provide a short explanation, choose from Waste, Road Damage, Water, Drainage, Energy, Public Safety, or Other, and attach the issue to a location.
- Multimodal AI analysis: the Gemini model considers the photograph and description together instead of treating them as unrelated inputs. The result includes issue classification, severity, environmental risk, public-safety risk, confidence, and a plain-language explanation.
- Transparent Priority Scoring Engine: a 0–100 score combines severity, environmental risk, public risk, location sensitivity, and recurrence or cluster weighting. The important point is that the score can be explained instead of appearing as an unexplained AI decision.
- Automatic hotspot clustering: nearby or repeated reports can be grouped so officers can identify recurring areas that deserve a broader intervention.
- Interactive geographic map: the Leaflet-based map shows reports and clustered hotspots so patterns can be understood by location as well as by category.
- Municipal command dashboard: officers can see a high-level health score, critical issues, high-priority issues, active reports, and resolved cases, while tracking report states from New through Under Review, Action Recommended, and Resolved.
- AI-generated insights and prediction: current reports can be synthesised into higher-level patterns and future-risk scenarios, helping officers move from reacting to problems toward preparing for them.
- Deterministic fallback scoring: when the live AI service cannot be reached, a rules-based engine keeps the main scoring workflow available. This reduces the chance that a temporary API issue will make the entire prototype unusable.

3.3 Innovation and Uniqueness

The main difference from a basic complaint-logging system is what happens after submission. VÉQUALUNE CIVIC combines multimodal analysis, explainable scoring, geographic clustering, dashboards, and predictive insights so each report contributes to a wider operational picture. Explainability is especially important. A municipal officer may need to justify why one issue was treated as urgent while another was scheduled later. For that reason, the platform presents a visible scoring breakdown rather than hiding the reasoning behind a single AI label. Hotspot detection adds another useful layer because repeated reports from the same area may be more important than any one report viewed alone. In the Sri Lankan context described by this proposal, this moves the system from passive record-keeping toward more proactive civic planning.

3.4 System Architecture

The platform follows a client-server architecture with three logical layers that keep responsibilities clear.

- **Presentation Layer** — a React 19 and TypeScript single-page application, built with Vite, provides the citizen reporting interface, map, command dashboard, insight views, and prediction views.
 - **Application/API Layer** — a Node.js and Express server provides REST endpoints for submitting reports, requesting AI analysis, retrieving hotspots, loading community metrics, and generating insights.
 - **Intelligence Layer** — the Gemini API provides multimodal image-and-text analysis, while the deterministic scoring engine provides a consistent fallback and supports auditable priority calculations.
- The screenshot set supplied with this proposal demonstrates how these layers appear to a user. The landing screen introduces the civic intelligence idea, the AI workflow accepts evidence, the analysis screen shows processing steps, and the resulting assessment turns a raw photograph into a category, risk view, score, explanation, and recommended action. The later map, command, insight, and prediction screens show how the same information can be reused for wider operational planning.

3.5 Data Flow

1. A citizen submits a report through the Report page with an image, description, category, and location.
2. The Express backend receives the submission and, when a valid Gemini API key is configured, sends the image and description to the Gemini model for structured analysis.
3. The scoring engine uses the AI assessment, or the fallback rules when necessary, to calculate a weighted Priority Score.
4. The report is stored with its scoring details and compared with existing records for geographic clustering and hotspot detection.
5. The dashboard, map, and insight views query the backend for aggregated data, making the processed information visible to municipal users in a form that supports action.

3.6 Technology Justification

React, TypeScript, and Vite are well suited to a fast-moving prototype because they support reusable components, clear type checking, and quick iteration. They allow the reporting form, dashboards, cards, filters, and map views to share a consistent interface without making the application unnecessarily heavy.

Node.js and Express provide a lightweight backend that is easy for a student development team to understand and extend. They are also a practical fit for REST APIs and server-side integration with Gemini. Leaflet is used for the geographic view because it is open source, lightweight, and appropriate for showing geolocated reports and hotspot clusters.

Gemini is important because the platform needs to understand both the image and the written description. The prototype can ask for structured output, which allows the result to connect directly to the scoring engine and user interface. The deterministic fallback is equally important because a civic system should degrade gracefully instead of stopping completely when an external AI service is unavailable.

3.7 AI Models

The platform uses Google Gemini through the `@google/genai` SDK in two main ways. The first is report-level analysis: the model examines the supplied photograph and description and returns structured information such as category, severity, risk levels, confidence, hazard tags, and a recommended response. The second is community-level synthesis: the model reviews the current set of reports and helps generate broader insights and predictive scenarios for the municipal dashboard.

Structured JSON output is requested so the AI result can move into the application in a predictable format. This matters because the next step is not a conversation with a chatbot; it is a scoring and decision-support pipeline. The AI output must therefore be consistent enough to feed the user interface and the scoring logic.

3.8 Security Considerations

- API keys such as GEMINI_API_KEY remain on the server and are managed through environment variables rather than being exposed in the client application.
- Report submissions are validated for fields such as description length, selected category, and coordinate bounds to reduce malformed or unexpected input.
- The fallback engine provides graceful degradation when the AI service is unreachable or incorrectly configured.
- For a production version, the proposal plans authenticated municipal accounts, rate limiting to reduce spam and abuse, and stronger image-content validation before storage.

The prototype also presents a responsible-AI position: AI recommendations are decision support, not automatic government decisions. The database architecture screen included in the supplied screenshots describes zero storage of personal identifying information in public intelligence views, server-side image processing with metadata scrubbing before storage, and human sign-off for AI recommendations. These principles help keep the system focused on community infrastructure rather than unnecessary personal data.

4. Target Audience

4.1 Primary Users

- Citizens — residents who notice a civic problem and need a simple way to report what they see from their phone or computer.
- Municipal and local-authority staff — officers who review reports, compare priorities, inspect hotspot patterns, and decide what field action is needed.

4.2 Beneficiaries

The wider community benefits when high-risk problems are identified earlier, repeated trouble spots receive more attention, and public infrastructure is maintained more consistently. Local government bodies benefit from a clearer evidence base for maintenance planning, resource allocation, and follow-up. Over time, the same records can help show which interventions are working and where problems continue to return.

4.3 Stakeholders

- Urban Councils and Pradeshiya Sabhas responsible for local infrastructure and public-service maintenance.
- Community organisations and school civic clubs that can encourage responsible reporting and public participation.
- Future integration partners such as water and electricity service providers, which could receive relevant reports through cross-agency routing.

5. Technical & Business Feasibility

5.1 Technical Feasibility

The core stack is technically feasible for the project because it is already represented in a working prototype. The current system demonstrates report submission, AI analysis, priority scoring, hotspot mapping, dashboards, and predictive insight modules. The fallback scoring engine further reduces technical risk because the central idea of transparent prioritisation does not depend entirely on a live external model at the exact moment of demonstration.

The prototype also supports staged development. The team can strengthen persistence, authentication, deployment, and integration later without changing the core idea, making the project realistic beyond the competition.

5.2 Resource Requirements

- **Development:** a three-member student team can divide responsibilities across the front end, backend/API work, and AI integration while working collaboratively on testing and presentation.
- **Infrastructure:** a Gemini API key and a hosting platform such as Vercel or Render are enough for the prototype and demonstration scale described in this proposal.
- **Data:** realistic sample reports covering the six main issue categories and Sri Lankan-style locations are needed so that the dashboard, map, and predictive modules can be demonstrated meaningfully.

5.3 Potential Challenges

- **API limits or quota exhaustion** during a live demonstration. The fallback scoring engine reduces this risk.
- **Limited real-world report volume** at launch. The prototype can use realistic seeded data during evaluation while keeping a clear path to genuine citizen-generated reports later.
- **Location accuracy** on lower-end devices. The reporting flow can allow a user to correct the suggested location manually.
- **AI response latency.** Loading states and the fallback engine help keep the interface understandable while processing takes place.

6. Development Roadmap

6.1 Proposed Development Plan

The development plan follows an iterative approach. The working prototype already covers the main end-to-end journey, so the next focus is refinement, testing, and production readiness rather than building the idea from zero. This is important for a competition project because a stable and understandable prototype is more useful than a long list of partially completed features.

6.2 System Architecture Overview

Technology Stack & Tools

- Programming Languages: TypeScript and JavaScript.
- Frameworks and Libraries: React 19, Vite, Express, Leaflet, Lucide icons, and Motion for interface animation.
- Data Storage: structured JSON or sample data in the current prototype, with planned migration to a persistent database such as PostgreSQL for production.
- APIs and Third-Party Services: Google Gemini API through @google/genai for multimodal analysis.
- Cloud Platforms and Hosting: Vercel for the front end or a Node-compatible host such as Render for the Express backend.
- AI/ML Technologies: Gemini multimodal analysis plus a custom deterministic scoring algorithm as a rules-based complement.
- Hardware: no special hardware is required beyond a standard smartphone or computer with a camera and internet access.

6.3 Key Milestones

The current milestone plan reflects the state of the working prototype and the remaining work needed to move toward a deployable system.

Table 1: Key Development Milestones

Milestone	Target Timeframe	Deliverable
Core prototype (reporting, AI analysis, scoring)	Completed (pre-submission)	Working end-to-end report → AI analysis → priority score flow
Hotspot clustering & map view	Completed (pre-submission)	Interactive map with live reports and clustered hotspots
Dashboard & predictive insights	Completed (pre-submission)	Municipal dashboard with metrics and AI-generated scenarios
UI polish, sample data & demo hardening	Panel Discussion stage	Refined UI, richer demo dataset, presentation-ready build
Authentication, persistence & production hosting	Post-panel / final stage	Staff login, persistent database, public deployment

6.4 Expected Deliverables

- A fully functional deployed web application demonstrating the citizen-reporting-to-municipal-dashboard workflow.
- Documented API endpoints for report submission, AI analysis, hotspot retrieval, and insight generation.
- A presentation-ready demo dataset covering all six main issue categories across sample Sri Lankan locations.
- A polished interface and screenshot-based demonstration that clearly explains what happens from report submission to municipal action.

6.5 Potential Technical Challenges

As the number of reports grows, the hotspot-clustering logic will need to scale efficiently. The planned direction is spatial indexing, such as geohashing, so that nearby reports can be identified without repeatedly comparing every report with every other report. This is a natural production-stage improvement and does not block the current prototype.

7. Sustainability and Future Enhancements

7.1 Long-Term Viability

Civic infrastructure reporting is a recurring need, so the platform has a natural long-term use case. A council does not need to solve only one problem once. Waste, drainage, roads, water, energy, and public-safety issues continue to appear, and their patterns can change over time. Once the platform is adopted by even one local authority, the value of the historical dataset can grow because new reports can be compared with earlier incidents.

The long-term value is not only faster complaint handling. Historical reports can reveal recurring hotspots and support preventive action instead of repeated one-off repairs.

7.2 Scalability

- Horizontal scalability: one pilot council can be extended to multiple municipalities, each with its own operational view and data boundaries.
- Data scalability: a production database such as PostgreSQL with PostGIS can support larger report volumes and spatial queries as adoption increases.

7.3 Future Improvements

- A native React Native mobile application for easier reporting while users are in the field.
- SMS or WhatsApp reporting for citizens who have limited access to a full web application, improving digital inclusion.
- Sinhala and Tamil interfaces so the service can be more accessible across Sri Lankan communities.
- Direct links to municipal work-order systems so high-priority issues can move from recommendation to field-team assignment more quickly.
- A public transparency portal using anonymised, aggregate statistics to show resolution progress without exposing personal information.

These improvements are deliberately practical. They extend access, improve operational integration, and strengthen trust without changing the core principle of the platform: turn community observations into information that supports better action.

8. Details of all team members



Full Name : **Morahelage Sinura Damsath Perera**

Role : Team Lead & Full-Stack Developer

Email Address : sinuradamsath2022@gmail.com

Mobile Number : +94 77 447 0098



Full Name : **Kodithuwakku Arachchilage Vihas Dinsitha**

Kodithuwakku

Role : Frontend Developer & UI/UX Designer

Email Address : iakodithuwakku0101@gmail.com

Mobile Number : +94 77 354 9707



Full Name : **Jayasooriya Mudiyanseelage Gangoda**

Walauwe Dulnith Sanvidu Bandara

Role : AI Integration & Backend Developer

Email Address : bandaradulnith@gmail.com

Mobile Number : +94 71 326 4178

9. Sketches, Diagrams, Mockups or Wireframes

9.1 High-Level System Architecture

Citizen Web App → Express API Server → Gemini API (AI Analysis) + Deterministic Scoring Engine → Report Store → Hotspot Clustering → Dashboard / Map / Insights for Municipal Staff.

This flow summarises the heart of VÉQALUNE CIVIC. The citizen starts with an observation. The backend converts it into structured information. The intelligence layer evaluates it. The scoring and clustering stages add operational context. Finally, the dashboard, map, insight, and prediction views turn that processed data into something a municipal team can use. The screenshots below document the working prototype and show the major screens in this journey.

9.2 Working Prototype Screenshots

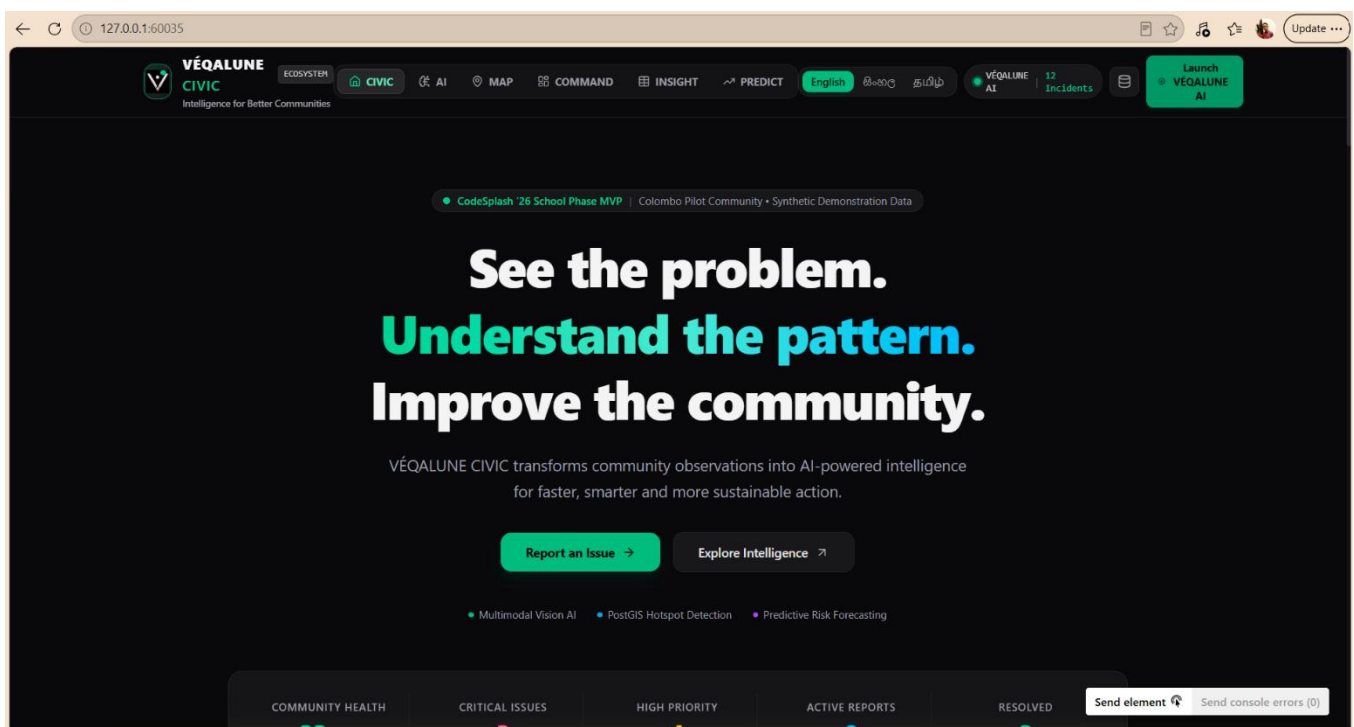


Figure 1. VÉQALUNE CIVIC landing page.

The landing screen introduces the civic intelligence concept and the main modules of the platform.

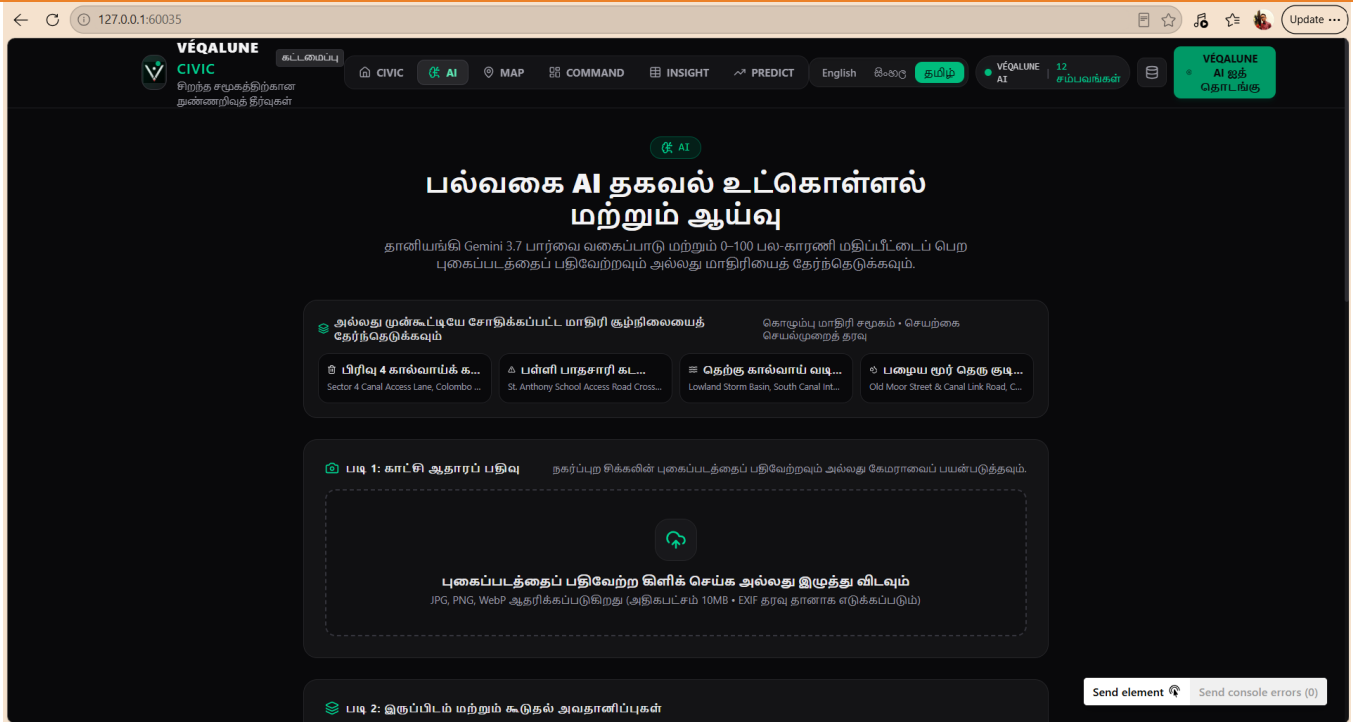


Figure 2. AI observation intake screen.

The intake screen demonstrates evidence upload, quick test presets, and location-aware reporting.

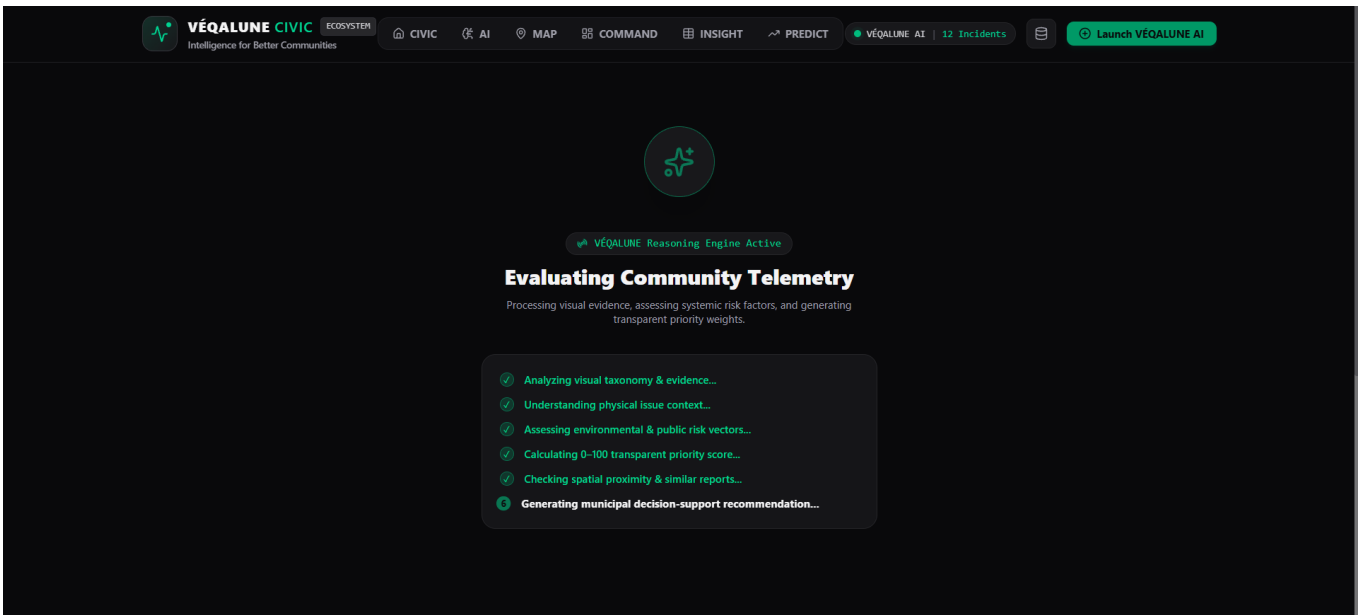


Figure 3. AI reasoning and evaluation pipeline.

The processing screen shows the main reasoning and scoring steps before an issue becomes a recommendation.

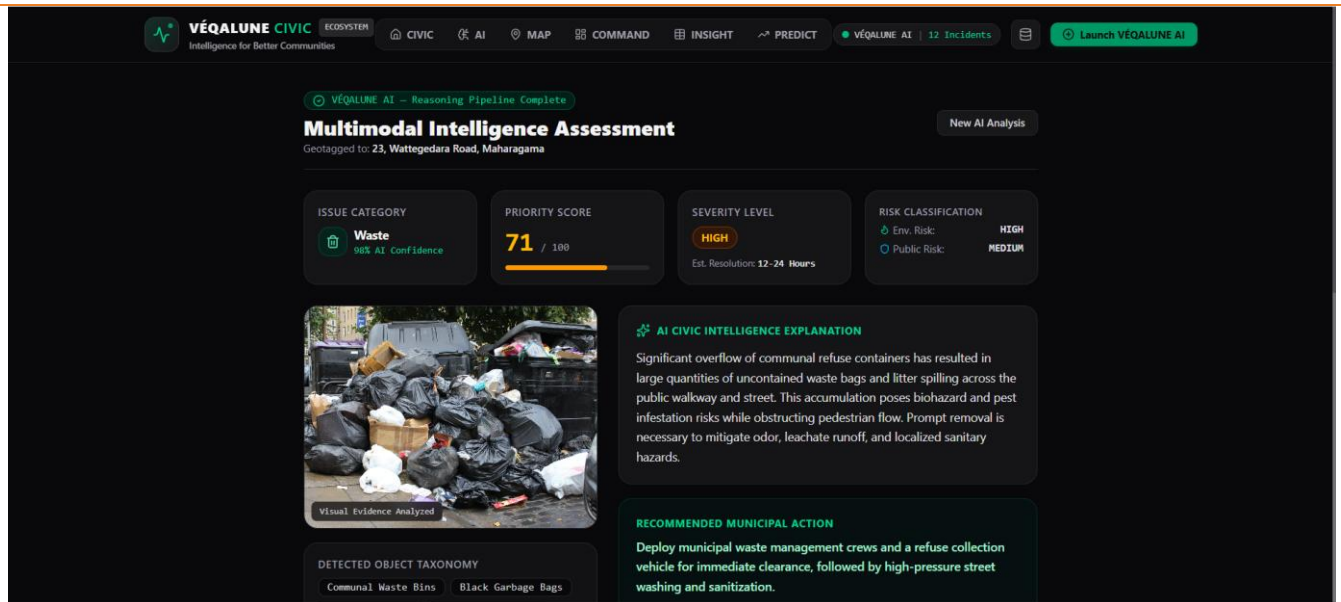


Figure 4. Multimodal intelligence assessment.

The assessment view displays category, AI confidence, Priority Score, risk levels, explanation, detected objects, and recommended action.

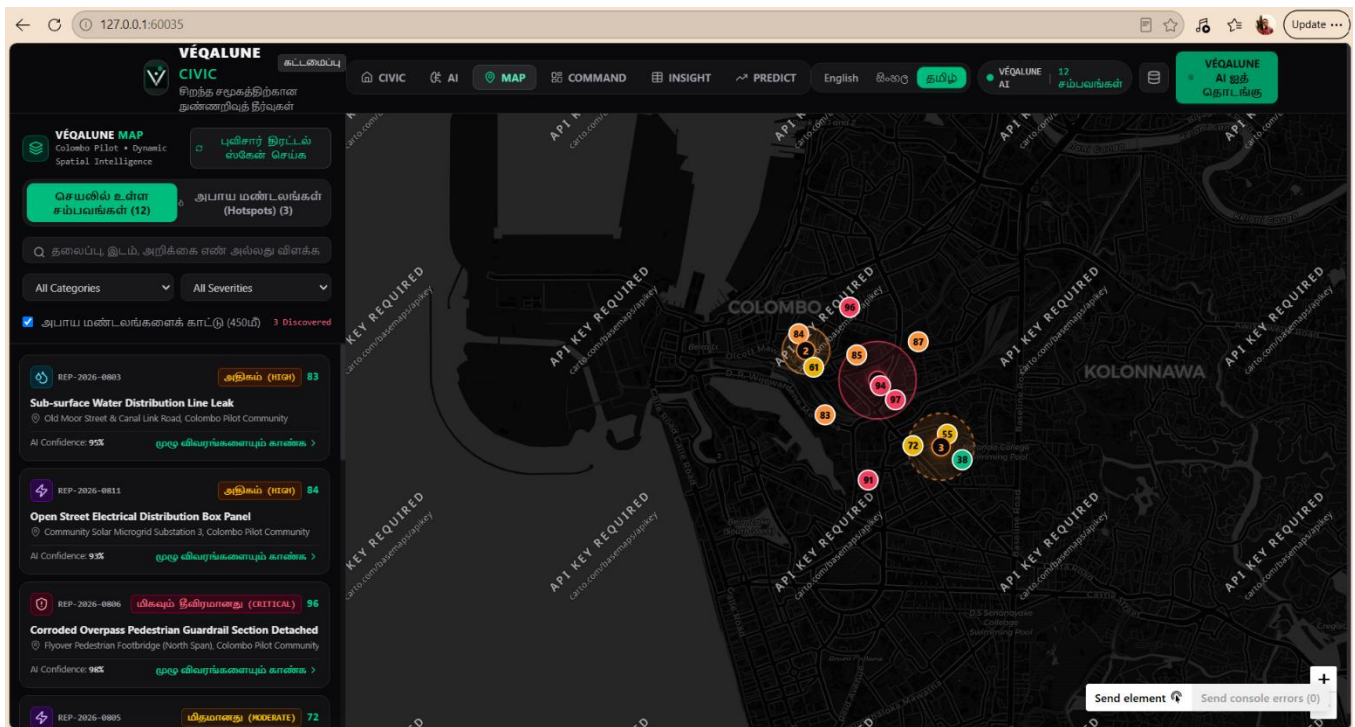


Figure 5. VÉQALUNE MAP.

The map view plots reports, scores, filters, and hotspot buffers to make geographic patterns visible.

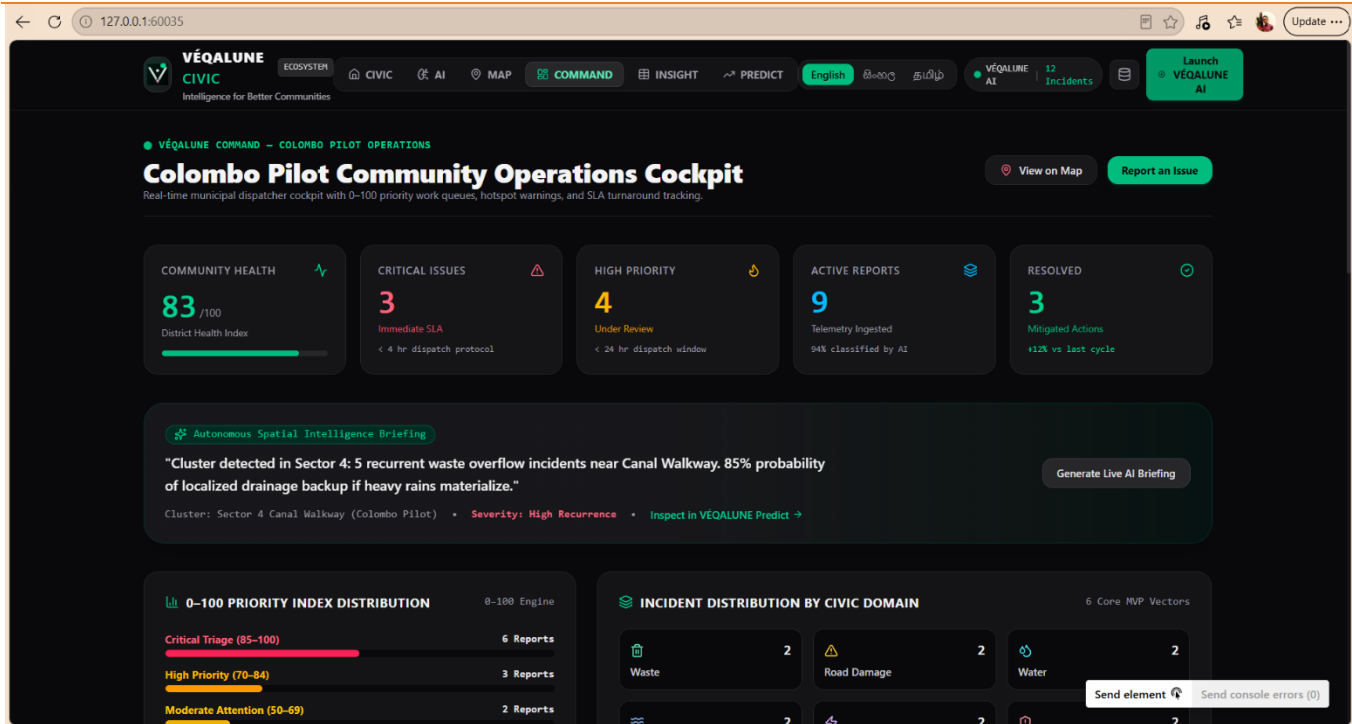


Figure 6. District Operational Command Center.

The command dashboard summarizes health, critical issues, priorities, active reports, resolved cases, and emerging patterns.

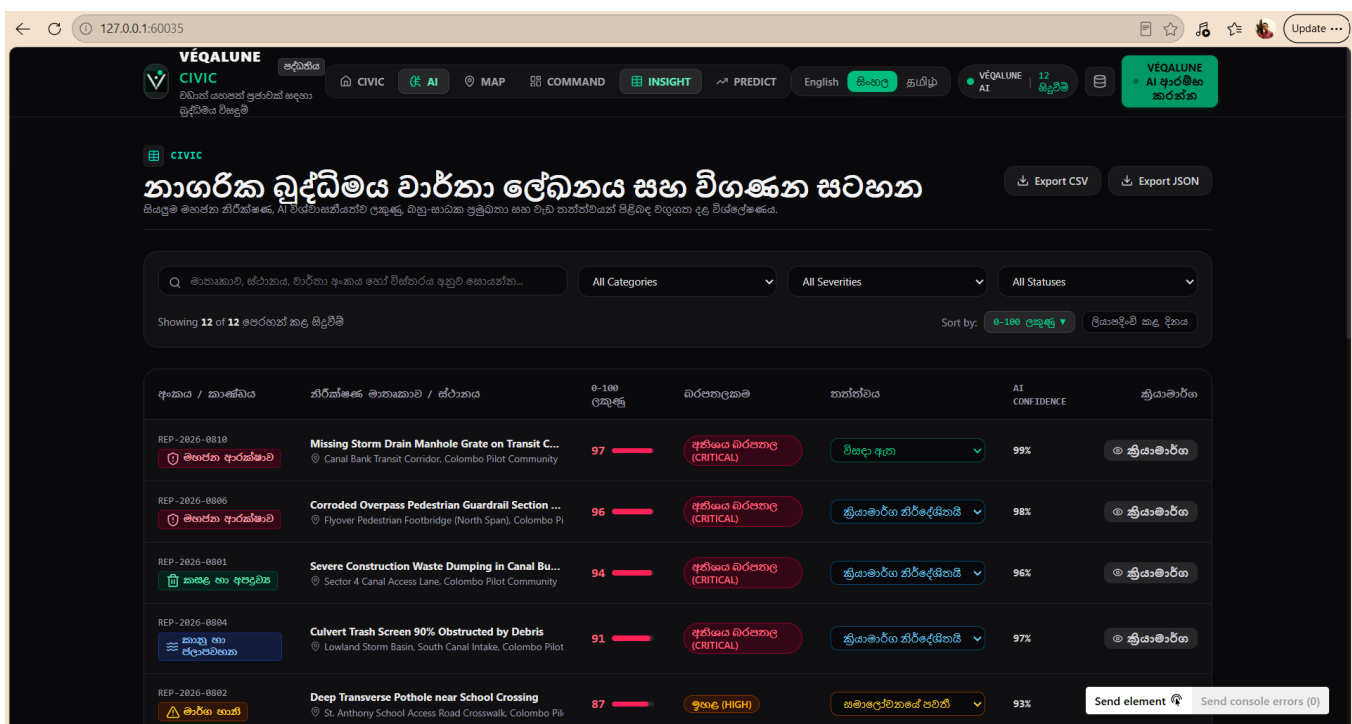


Figure 7. Civic incidents and systemic patterns.

The insight screen supports searching and filtering incidents by location, priority, severity, status, and AI confidence.

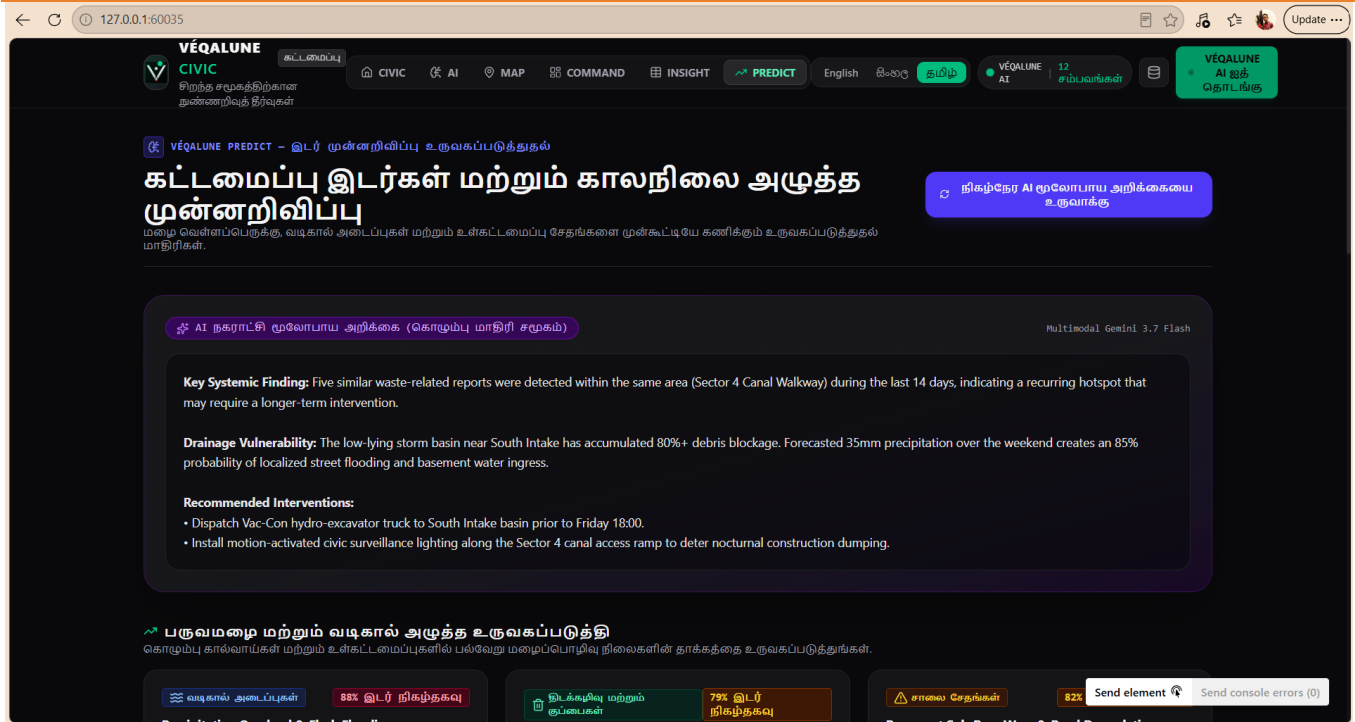


Figure 8. Systemic risk and climate stress forecasting.

The prediction screen uses scenario modelling to highlight recurring waste, drainage vulnerability, rainfall risk, and interventions.

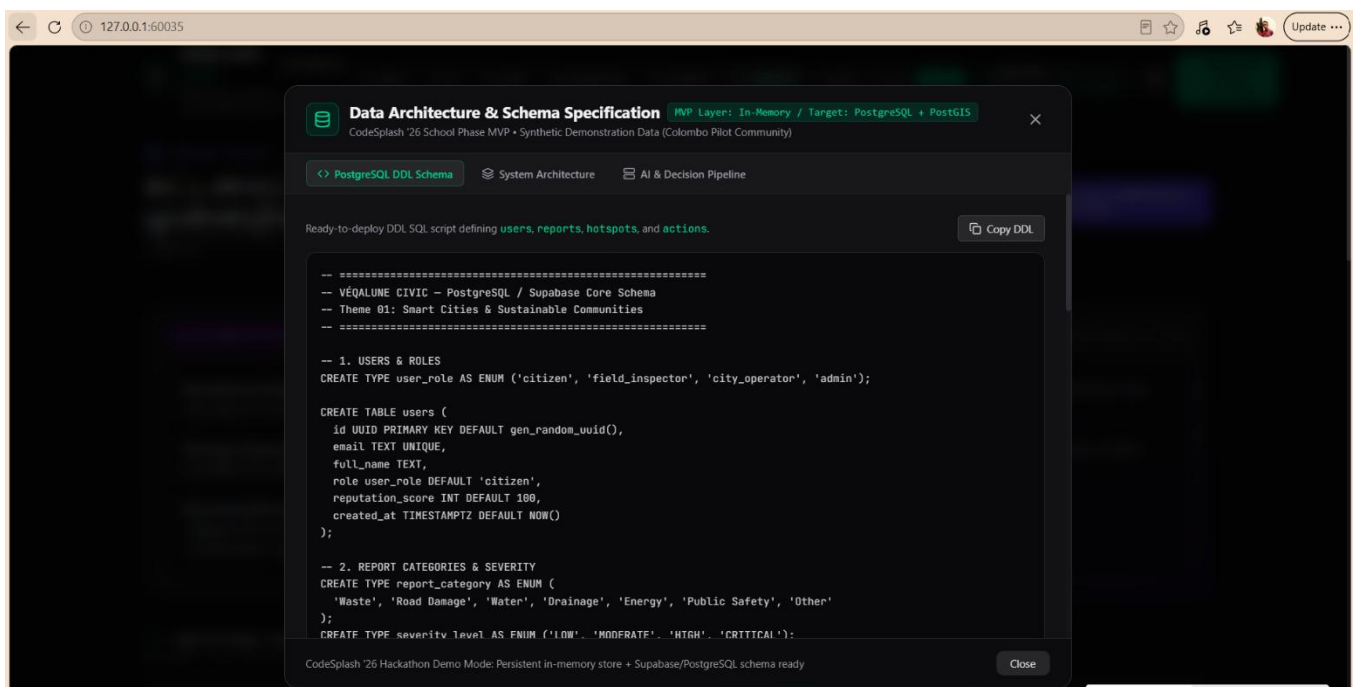


Figure 9. Database architecture and responsible-AI framework.

The architecture screen maps the six modules and records the prototype approach to privacy, metadata scrubbing, and human sign-off.

10. References

[1] Live Veqalune-Civic Platform

<https://veqalune-civic.vercel.app>

[2] Brand Identity of Veqalune-Civic

<https://drive.google.com/drive/folders/1qPBbFLikWXxCNEQWPWH93iIckdFLyfB2?usp=sharing>

[3] Promotional Video of Veqalune-Civic

https://drive.google.com/file/d/1sKDZtQWStPz2W2M4r6djzpRse_XJ8lEa/view?usp=sharing